

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 5. Conditional Statements

Lecture: 43. Challenge - Discounted Bill

Section 1: Lecture Summary

This lecture presents a programming challenge to calculate a discounted bill amount based on the original bill amount using Python's conditional statements (if-elif-else). The main idea is to apply different discount rates depending on the bill amount thresholds. Students are guided through the logic of the challenge, which involves:

- Determining the discount percentage based on the bill amount bracket.
- Computing the discounted price by subtracting the discount from the original amount.
- Writing a Python program that uses an if-elif-else ladder to select the correct discount.

The lecture explains the conditions for each discount bracket:

- Less than 1000 → 10% discount
- From 1000 (inclusive) to less than 5000 → 15% discount
- From 5000 (inclusive) to less than 10000 → 20% discount
- 10000 and above → 25% discount

The instructor encourages students to write and try this program in any Python environment such as PyCharm or Jupyter Notebook, reinforcing the use of variables, typecasting inputs, and conditional branching. The lecture concludes by promising more challenges in future sessions.

Section 2: Key Concepts and Explanations

- Conditional Statements (if-elif-else ladder):

- Used to execute certain blocks of code based on boolean conditions.

- ``if`` tests the first condition.
- ``elif`` tests additional conditions sequentially.
- ``else`` handles all remaining cases if none of the previous conditions is met.

- **Comparison Operators:**

- ``<`` (less than), ``>=`` (greater than or equal to), and logical ``and`` to combine multiple conditions.

- **Discount Calculation:**

- Discount amount is calculated as a percentage of the original bill.
- Formula: ``discount_amount = amount * discount_rate``
- Discounted total = ``amount - discount_amount``
- The discount rates used are: 10% (0.1), 15% (0.15), 20% (0.2), 25% (0.25).

- **Variables and Input:**

- ``amount`` variable holds the original bill value.
- Input can be taken from the user and typecast to integer or float before calculations.

- **Program Flow Example:**

- Check if the amount is less than 1000 → apply 10% discount.
- Else if amount is between 1000 and less than 5000 → apply 15% discount.
- Else if amount is between 5000 and less than 10000 → apply 20% discount.
- Otherwise (amount $\geq$ 10000) → apply 25% discount.

Section 3: Example Code and Use Cases

Take input from user and convert to integer

Calculate total amount after discount based on slabs

```
amount = int(input("Enter Bill Amount: "))

if amount < 1000:
    total = amount - amount * 0.10 # 10% discount
elif amount >= 1000 and amount < 5000:
    total = amount - amount * 0.15 # 15% discount
elif amount >= 5000 and amount < 10000:
    total = amount - amount * 0.20 # 20% discount
else:
    total = amount - amount * 0.25 # 25% discount

print("Total amount to pay after discount:", total)
```

- Explanation:

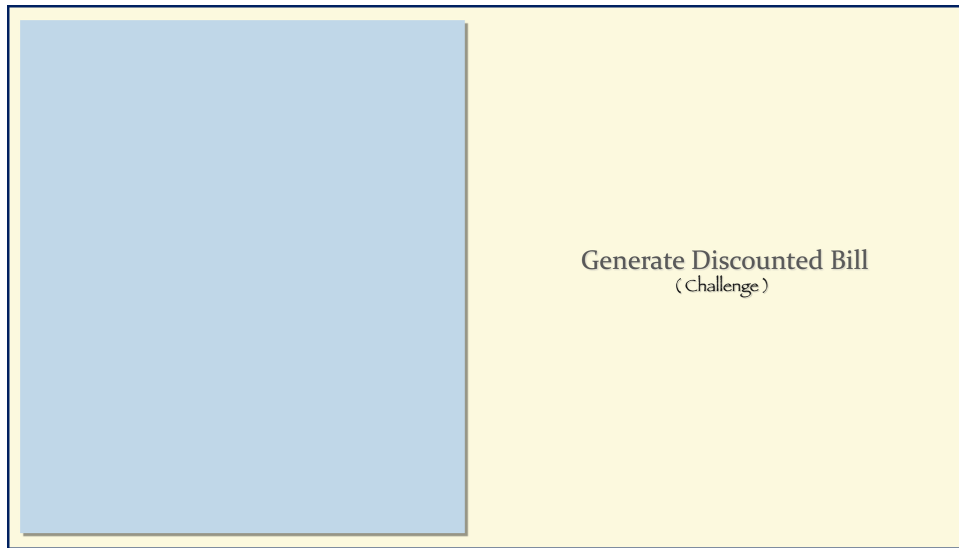
This program reads the bill amount, checks which discount bracket it fits into, calculates the discounted amount using percentage multiplications, and then prints the final payable amount.

Section 4: Key Takeaways

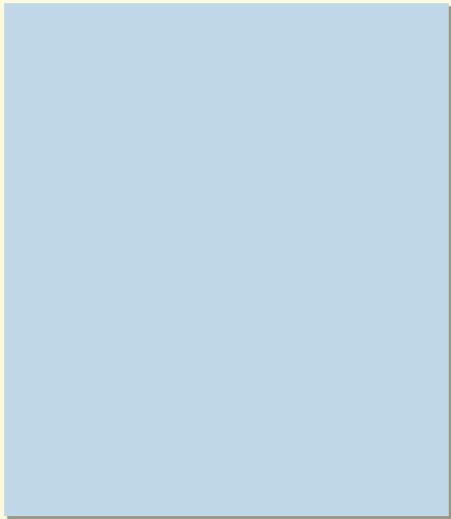
- Conditional statements (`if-elif-else`) allow branching logic based on multiple conditions efficiently.
- Discounts can be implemented using simple arithmetic operations combined with conditional logic.
- Careful use of comparison operators and logical conditions ensures correctness of discount application.
- Typecasting inputs helps avoid errors during numerical calculations.
- Breaking down the problem into defined slabs makes coding more organized and maintainable.
- The else clause handles all values not covered by prior conditions, simplifying the logic.
- This challenge illustrates combining decision-making with basic math operations in Python.

Lecture in Slides

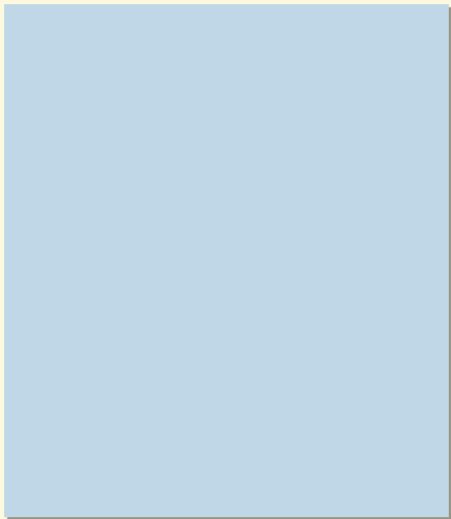
Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Generate Discounted Bill
(Challenge)



Generate Discounted Bill
(Challenge)



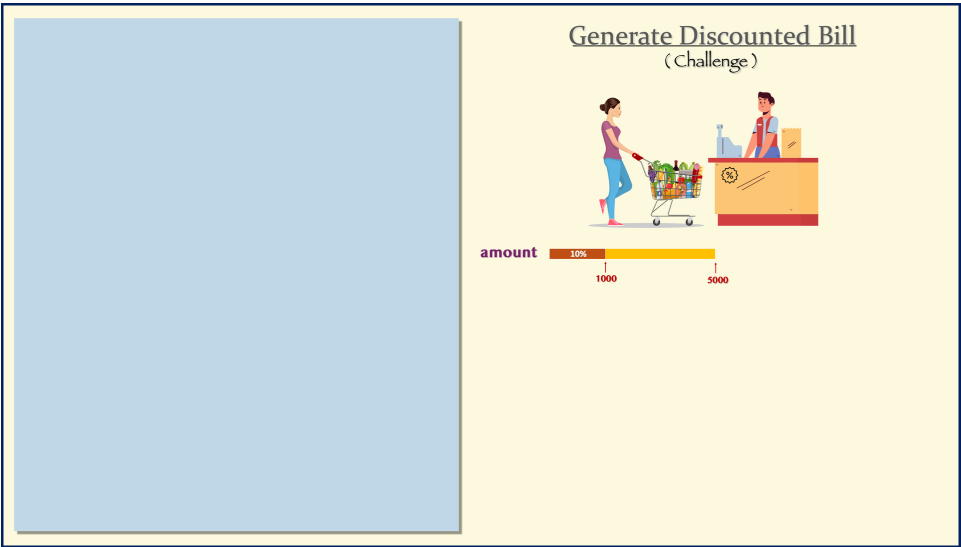
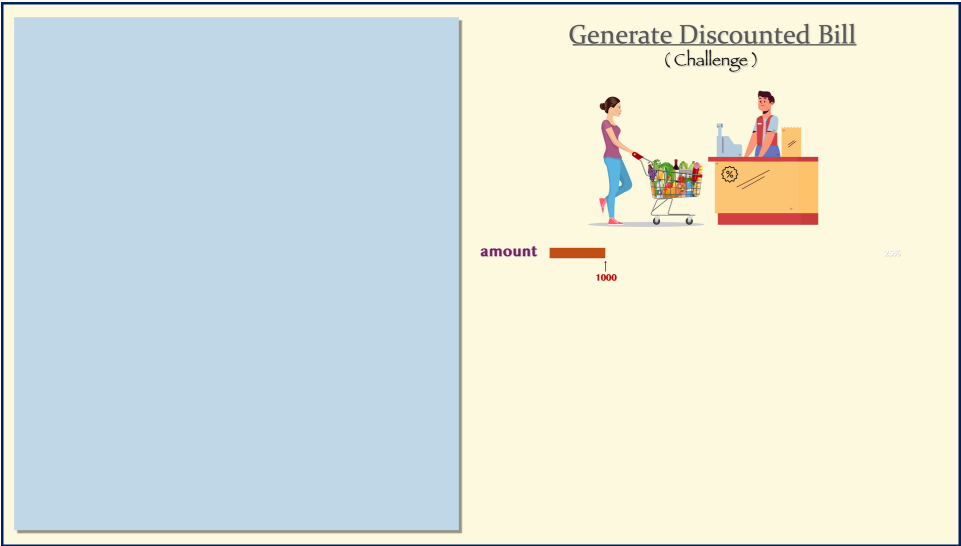
Generate Discounted Bill
(Challenge)

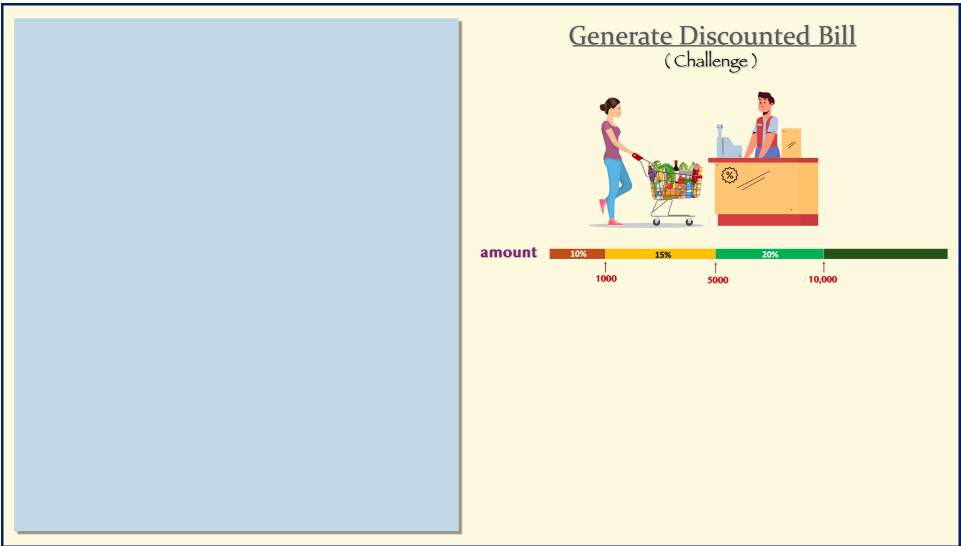
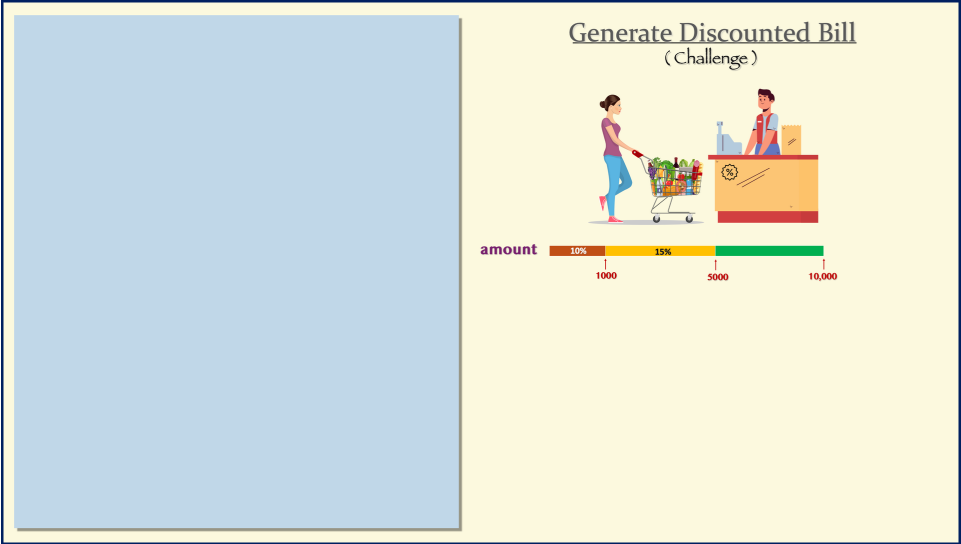


Generate Discounted Bill
(Challenge)



amount





Generate Discounted Bill

(Challenge)



amount

10%	15%	20%	25%
1000	5000	10,000	

10%

Generate Discounted Bill

(Challenge)



amount

10%	15%	20%	25%
1000	5000	10,000	

10%

amount =

Generate Discounted Bill
(Challenge)



10%

amount = amount

Generate Discounted Bill (Challenge)



10%

amount = amount - amount * 0.1

Generate Discounted Bill (Challenge)

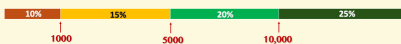


amount = amount - amount

Generate Discounted Bill

(Challenge)



amount 

15%

amount = amount - amount

Generate Discounted Bill

(Challenge)



amount 

Generate Discounted Bill

(Challenge)



amount

10%	15%	20%	25%
1000	5000	10,000	

Generate Discounted Bill

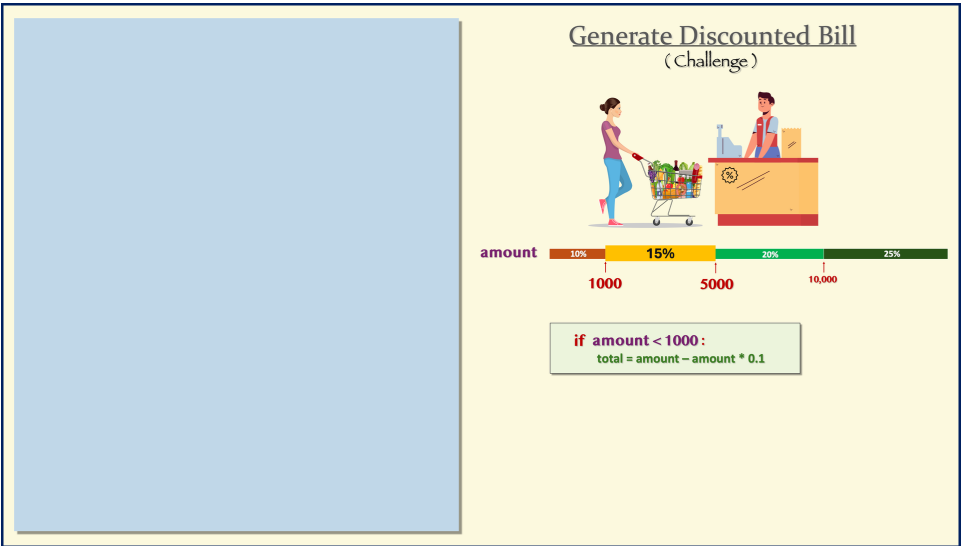
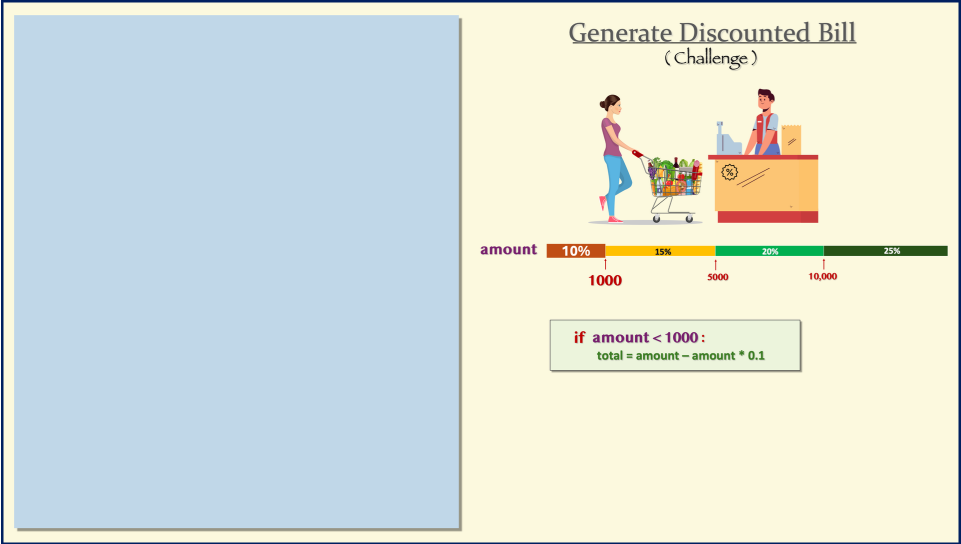
(Challenge)



amount

10%	15%	20%	25%
1000	5000	10,000	

if amount < 1000 :



Generate Discounted Bill

(Challenge)



amount

10%	15%	20%	25%
1000	5000	10,000	

```
if amount < 1000:  
    total = amount - amount * 0.1  
elif amount >= 1000 and amount < 5000:
```

Generate Discounted Bill

(Challenge)



amount

10%	15%	20%	25%
1000	5000	10,000	

```
if amount < 1000:  
    total = amount - amount * 0.1  
elif amount >= 1000 and amount < 5000:  
    total = amount - amount * 0.15
```

Generate Discounted Bill
(Challenge)



```
if amount < 1000:  
    total = amount - amount * 0.1  
elif amount >= 1000 and amount < 5000:  
    total = amount - amount * 0.15
```

Generate Discounted Bill (Challenge)



```
if amount < 1000:  
    total = amount - amount * 0.1  
elif amount >= 1000 and amount < 5000:  
    total = amount - amount * 0.15  
elif amount >= 5000 and amount < 10000:
```

Generate Discounted Bill (Challenge)



```
if amount < 1000:  
    total = amount - amount * 0.1  
elif amount >= 1000 and amount < 5000:  
    total = amount - amount * 0.15  
elif amount >= 5000 and amount < 10000:  
    total = amount - amount * 0.2
```

Generate Discounted Bill

(Challenge)



amount

10%	15%	20%	25%
1000	5000	10,000	

```
if amount < 1000:
    total = amount - amount * 0.1
elif amount >= 1000 and amount < 5000:
    total = amount - amount * 0.15
elif amount >= 5000 and amount < 10000:
    total = amount - amount * 0.2
```

Generate Discounted Bill

(Challenge)



amount

10%	15%	20%	25%
1000	5000	10,000	

```
if amount < 1000:
    total = amount - amount * 0.1
elif amount >= 1000 and amount < 5000:
    total = amount - amount * 0.15
elif amount >= 5000 and amount < 10000:
    total = amount - amount * 0.2
else:
```

Generate Discounted Bill (Challenge)



```
if amount < 1000:  
    total = amount - amount * 0.1  
elif amount >= 1000 and amount < 5000:  
    total = amount - amount * 0.15  
elif amount >= 5000 and amount < 10000:  
    total = amount - amount * 0.2  
else:  
    total = amount - amount * 0.25
```

Generate Discounted Bill (Challenge)



```
if amount < 1000:  
    total = amount - amount * 0.1  
elif amount >= 1000 and amount < 5000:  
    total = amount - amount * 0.15  
elif amount >= 5000 and amount < 10000:  
    total = amount - amount * 0.2  
else:  
    total = amount - amount * 0.25  
print('Pay:', total)
```


Generate Discounted Bill

(Challenge)



amount

amount	10%	15%	20%	25%
1000				
5000				
10,000				

```
if amount < 1000:
    total = amount - amount * 0.1
elif amount >= 1000 and amount < 5000:
    total = amount - amount * 0.15
elif amount >= 5000 and amount < 10000:
    total = amount - amount * 0.2
else:
    total = amount - amount * 0.25
print("Pay: ", total)
```

Generate Discounted Bill

(Challenge)



amount

amount	10%	15%	20%	25%
1000				
5000				
10,000				

```
amount = ?
if amount < 1000:
    total = amount - amount * 0.1
elif amount >= 1000 and amount < 5000:
    total = amount - amount * 0.15
elif amount >= 5000 and amount < 10000:
    total = amount - amount * 0.2
else:
    total = amount - amount * 0.25
print("Pay: ", total)
```

Generate Discounted Bill (Challenge)



```
amount = int( input( 'Enter Bill Amount' ) )  
  
if amount < 1000:  
    total = amount - amount * 0.1  
elif amount >= 1000 and amount < 5000:  
    total = amount - amount * 0.15  
elif amount >= 5000 and amount < 10000:  
    total = amount - amount * 0.2  
else:  
    total = amount - amount * 0.25  
  
print( 'Pay: ', total )
```